

"Likelihood Ratio" for WVN

⇒

$$LR = \frac{P(\bar{y} | H_1)}{P(\bar{y} | H_0)}$$

for WVN

$$P(y_i | H_1) = \begin{cases} 1 & \text{if } y_i \in [\mu - 0.5, \mu + 0.5] \\ 0 & \text{otherwise} \end{cases}$$

Similarly

$$P(y_i | H_0) = \begin{cases} 1 & \text{if } y_i \in [-0.5, 0.5] \\ 0 & \text{otherwise} \end{cases}$$

this leads to issues w/ our LR test (study by Zeng)

if $\max(\bar{y}) > 0.5$ or

$\min(\bar{y}) < -0.5$ we have

an indicator our data is wrong.

We want LR to check the most likely value of μ . We can do this by looking at the midpoint. This is better than averages b/c it works well with the rules for our system. μ can best be found when looking at our range of values.

best LR

$$\hat{\mu}_{(3)} = \frac{\min(\bar{y}) + \max(\bar{y})}{2}$$